ORGANISING ADHD USERS' THOUGHTS WITH A MULTI-TASK FLAN-T5 MODEL

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ABSTRACT

Attention-Deficit Hyperactivity Disorder (ADHD) is a condition that affects cognitive functions such as the ability to concentrate and organise thoughts. Individuals often struggle with scattered and disorganised thinking, impacting their daily lives and productivity. ADHD has been recognised as a disability and is affecting more individuals worldwide. While there are traditional approaches such as medication, behavioural therapy with medical professionals' guidance to ease the symptoms, the availability of tools to help them articulate their thoughts is rare. This paper presents the development of a fine-tuned FLAN-T5 model to assist ADHD individuals in classifying, rewriting, and summarising their thoughts, aiming to assist them in navigating daily life. The study utilised the comments and posts from the largest Reddit community r/ADHD, a dataset spanning from 2008 to 2024. A FLAN-T5-Base model was trained and fine-tuned using the processed dataset to help the model learn ADHD-specific linguistic patterns and generate structured text output. The fine-tuned model achieved a micro-F1 of 0.5231 and macro-F1 of 0.4479 for the primary task, multi-label classification. While generation tasks showed low BLEU (0.1093 for rewrite) and ROUGE (0.3590 for summary), both tasks scored moderate BERTScore (0.4892 for summary, 0.4953 for rewrite), suggesting that the semantic meaning is preserved. The limitations include class imbalance, limited annotated examples, and the multi-label task complexity with mutual exclusivity rules. Possible future considerations include integrating the model into a web or mobile application, enabling user input for testing, and subsequent updates.

CCS CONCEPTS

Computing methodologies → Natural language processing

Computing methodologies → Neural networks

KEYWORDS

ADHD, multi-task learning, natural language processing, Flan-T5, text classification, text summarization, sentence rewriting.

1 INTRODUCTION

Attention-Deficit Hyperactivity Disorder (ADHD) is a common neurodevelopmental disorder, known by the characteristic of inattention, hyperactivity, and impulsivity that impact daily functioning and development. The disorder has found to affect approximately 5% of children globally, and 65% of them may persist the symptoms into adulthood (Chen, 2024). Research discovered that ADHD is often associated with learning disabilities, a higher risk of traffic accidents, addictions, engaging in crime, physical and mental health issues (Chen, 2024). Hutchison et al. (2016) found that preschool children with ADHD are more likely to exhibit thought disorder, having loose associations, bizarre ideas or illogical thinking. Similarly, Madiouni et al. (2020) discovered that adults with ADHD have a higher tendency to mind-wandering, with thoughts that are less structured and intentional.

Cognitive dysfunction of ADHD individuals often leads to procrastination, emotional regulations which impact their personal life, relationships and professional settings (Chen, 2024). Traditional treatments such as medication and behavioural interventions help manage symptoms. However, due to its heterogeneous nature, ongoing research continues to improve the methods of identification and diagnosis of ADHD in individuals. While these efforts enhance early diagnosis, the tools to help them navigate daily life without medical supervision remain limited. With the recent advancement in technology, particularly Natural Language Processing (NLP), machines can process and analyse human language by statistical features and models (Le Glaz et al., 2021). As NLP models have demonstrated decent results in statistical tasks like text classification, it is crucial to explore how NLP capabilities in classifying, rewriting and summarising will be helpful for ADHD users. Rather than relying on general dataset in common approach, a more tailored approach utilising ADHD dataset is needed to understand their linguistic patterns.

This research aims to fine-tune a FLAN-T5 model that will help ADHD individuals to structure their thoughts more effectively, reducing cognitive load and overwhelm, leading to enhanced productivity. It will serve as a proof-of-concept (POC) assistive tool to help them in classifying their random thoughts, rewriting text for clarity, and summarising the core idea to ease communication and daily challenges.

2 RELATED WORKS

2.1 ADHD AND COGNITIVE CHALLENGES

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder identified with inattention, hyperactivity, impulsivity, or a combination of these traits. While ADHD is mostly recognised and diagnosed in childhood, the diagnosis of adult ADHD remains challenging due to coexisting

mental health issues. Research reported that hyperactivity and impulsiveness may decline over the years, yet inattention and disorganisation may remain (Di Lorenzo et al., 2023).

One of the strengths of ADHD is divergent thinking, enabling individuals to produce multiple ideas and solutions simultaneously (Hoogman et al., 2020). While this can enhance creativity, it also contributes to excessive mind-wandering. Uncontrolled mind-wandering may lead to cognitive overload, cause overwhelming, and a higher risk of mental health issues such as anxiety and depression (Kandeğer et al., 2023; Yamaoka & Yukawa, 2020).

Research suggests that cognitive disorganisation develops early in ADHD and persists into adulthood. Hutchison et al. (2016) investigated the identification of thought disorder in preschoolers, comparing ADHD and control groups. The ADHD group shows three-time higher tendency to exhibit thought disorder (75% vs 25%), having unusual or dysfunctional ways of thinking. Thought disorder can be characterised by loose associations, bizarre ideas or illogical thinking, associated with long-term risk of social and cognitive dysfunction.

Correspondingly, Madiouni et al. (2020) investigated the mind-wandering and sleepiness between ADHD adult group and healthy control group. The results indicate that ADHD group show higher tendency for mind-wandering, mind-blanking, higher sleepiness, with thoughts that are less structured and intentional. These findings suggest that less structured thought processes persist in individuals with ADHD across their lifespan, posing challenges in daily life functioning, decision-making and productivity. Consequently, individuals with ADHD face significant challenges in real-life situations, including academic, personal life, career, and interpersonal relationships settings. They are associated with higher risk of lower education level, unemployment, underperformance in career, anti-social behaviour, and difficulties in building and maintaining relationships (Chen, 2024).

2.2 COMMUNICATION AND LANGUAGE CHALLENGES

While ADHD is mainly associated with hyperactivity, impulsivity, and inattention, language disturbances often occur. Language disturbance is defined as a type of communication disorder from structural and pragmatic aspects. Structural difficulties refer to problems with grammar, vocabulary, and phonology, while pragmatic difficulties refer to inappropriate use of language in social settings, such as conversation and interaction, affecting communication skills (Hawkins et al., 2016). Hawkins et al. (2016) conducted a study of 254 children with attention, learning, or memory difficulties, using parent-report questionnaires to assess ADHD symptoms and standardised tests to evaluate participants' language abilities. The results showed that core ADHD symptoms, particularly inattention and hyperactivity, are strongly associated with pragmatic language difficulties. These include interrupting conversations, going off-topic, talking excessively, and failing to interpret non-verbal cues. Conversely, structural language difficulties showed no significant link to ADHD behaviours. These issues may arise from underlying executive functioning failures, affecting working memory and inhibition systems, which may be caused by disruptions in neurodevelopmental systems related to cool cognitive-based functions and hot affective processes.

Population-level data from Liu et al. (2024) further support the links between ADHD and communication difficulties. Compared to non-ADHD peers, children with ADHD showed significantly lower scores in communication and general knowledge (mean = 1.86 versus 2.22; $t = -16.48$). In contrast, the difference in the ability to communicate adequately in the first language was relatively small (90.1% versus 92.4%; $t = 10.62$). These findings suggest that children with ADHD do not show major difficulties in structural language but are more likely to have problems in using language appropriately in daily communication.

2.3 SPEECH AND WRITING PATTERNS IN ADHD

Numerous studies have reported that individuals with ADHD show different patterns in writing compared to neurotypical peers. Re et al. (2007) conducted three studies in children from age 7 to 13 who showed ADHD symptoms, compared the results with a control group through a series of expressive writing tasks. Throughout three studies involved descriptive and narrative tasks using pictures, prompts, and a sequence of images, the results remained consistent. The outcome indicated that ADHD group wrote shorter texts with weaker structure and language ability, and generally made more errors in grammar, phonology, vocabulary and spelling.

Similarly, a number of studies have examined the speech of individuals with ADHD compared to neurotypical controls, assessing features such as tone, speed, fluency, and how clearly the content is connected and organised in conversation. Chan and Fugard (2018) reviewed 18 case notes of children referred for ADHD and found that 15 out of 18 showed signs of speech, language, and communication difficulties (SLCDs). However, only 4 children were referred for speech-language therapy, suggests that SLCDs may be underdiagnosed in ADHD children. Later, Etter et al. (2021) examined 28 adults with ADHD and 22 neurotypical adults aged between 18 to 26 years old. The results showed that adults with ADHD speak slower, and have weaker control over mouth pressure during speech. These features are linked to difficulties in speech motor planning and sensory feedback, which may affect fluency and clarity in communication. Recently, von Polier et al. (2025) utilised machine learning to analyse voice features of adults with ADHD. This research examined 920 recordings from 387 ADHD patients, 204 healthy controls, and 100 psychiatric controls aged 18-59 years. The results showed that pitch, rhythm, and loudness in speech could help predict ADHD with good accuracy (AUC = 0.77), especially in young females below 32 (AUC = 0.87). These findings suggest that speech differences in ADHD are not only observed by practitioners but are also detectable through acoustic features using machine learning techniques.

2.4 COGNITIVE LOAD AND BRAIN FOG

Cognitive overload may happen when the brain needs to handle too much information simultaneously, but the executive system cannot handle efficiently. Such overload may increase the mental effort, slow down the cognitive responses, leading to mental fatigue. Jureket al. (2025) found that individuals with ADHD commonly have difficulties in sensory processing, which increases distractibility and further burdens their cognitive functioning.

Fisher et al. (2023) carried out three experiments and found that people with higher ADHD symptoms performed worse under cognitive load but improved under perceptual load. fMRI results also showed lower brain network efficiency during cognitive load, which improved under perceptual load. Overall, these findings suggest that cognitive load worsens ADHD symptoms, while perceptual load may help support attention and brain function.
As a result of cognitive overload, many people report feeling "brain fog." While not a formal medical diagnosis, the term is commonly used to describe a sense of mental cloudiness. Denno et al. (2025) reviewed how brain fog appears in various health conditions to suggest a clearer definition. The report describes brain fog as a condition that includes slower thinking, difficulty focusing, memory problems, and language challenges. For ADHD individuals, this condition arises when excessive information overwhelming the executive functions.

2.5 MACHINE LEARNING APPLICATIONS IN ADHD

Machine Learning (ML) is a subset of Artificial Intelligence (AI), in which the system learns from data to improve the decision-making process. Natural Language Processing (NLP) is a subfield of computer science that enables machines to process, understand, and interpret human languages through analysing the patterns and relationships within text. Over the years, NLP has evolved from rule-based to statistical approaches, and integrated machine learning techniques such as neural networks to enhance performance recently and is commonly applied for text processing tasks such as text classification or sentiment mining (Le Glaz et al., 2021).
Using NLP-powered ML model, researchers have explored the potential of models in analysing linguistic patterns, behaviours and cognitive traits from ADHD group. Unlike traditional intervention that focusing on treatment with prescribed medication, socio-rehabilitation, psychotherapy, behavioural therapy to help manage their symptoms, machine learning could assist in early prediction, analysing and assessment.
As early intervention can reduce the harm of ADHD, researchers had employed the machine learning algorithm for early diagnosis among preschool children. Based on population-level surveillance data, the model delivers promising result, indicated that machine learning will be beneficial for early diagnosis of ADHD (Liu et al., 2024).
Additionally, Burger et al. (2021) explored the usage of NLP in extracting schema from patients' thought records, demonstrated promising result of identifying maladaptive thought patterns in cognitive therapy, comparative to manual assessment results. Thereafter, Shen et al. (2024) investigated the feasibility of using Large Language Model (LLM) to conduct Cognitive Behaviour Therapy, and the result is positive. Most recently, Karim et al. (2025) introduced a BERT-based model, "LastBERT" through knowledge distillation to classify the severity of concerns from ADHD users based on social media text data.
Despite these advancements, assistive technology tools remain largely unavailable for ADHD users. While professionals are getting more AI-driven options to assess, analyse and diagnose, the gap in directly assisting ADHD individuals, particularly in reducing cognitive load and thought organisation remains.

2.6 NLP FOR COMMUNICATION SUPPORT

Natural language processing (NLP) offers tools that may help improve clarity and structure, such as classification, rewriting, and summarisation. While classification helps find the topic or emotion of a message, rewrites help turn messy or disorganised text into clearer sentences, preserving the speaker's meaning. Summarisation makes long or scattered text shorter and more focused, while extracting the core idea from a lengthy text.
The Text-to-Text Transfer Transformer (T5) model by Google converts all tasks using the same method called text-to-text, which takes text as input to generate new text output. For example, it can take a sentence and support diverse tasks: a label for text classification, translating into another language, or a concise summary (Raffel et al.,2020). The T5 model has shown strong performance on diverse text-related tasks, showing comparable performance to task-specific architectures.
Later, Google launched another model to the T5 family, FLAN-T5 which added instruction tuning to the framework (Chung et al., 2024). FLAN-T5 was trained with massive text-to-text dataset just like T5, except that it was integrated using prompts like "classify this sentence" or "summarise the text." This made the model better at following instructions and able to perform well even on new tasks it had not seen before. Hence, the model is powerful and easier to use in real-world settings.
With these features, FLAN-T5 could support people with ADHD by helping them find the meaning or emotion in their message, rewrite unclear thoughts, and summarise their text. This shows good potential for building future tools that support communication for ADHD users.

2.7 RESEARCH GAP AND MOTIVATION

Although NLP has been used in ADHD diagnosis and assessment, research on using NLP to support users in communication is rare. Most studies focus on prediction or classification to help diagnose individuals with ADHD earlier. This study wants to fill that gap by building a lightweight NLP model trained on ADHD-related language from Reddit. The model is designed to support thought organisation by three tasks altogether: classification, rewriting, and summarisation. It aims to give practical support that matches the way ADHD users speak and write and help reduce difficulties in daily life.
Reddit posts often show informal, fragmented, and emotional language, which is common in people with ADHD. Previous research shows that this kind of data can be used to detect ADHD-related concerns, and smaller models can work well with such text (Karim et al., 2025). This study employs the FLAN-T5 model, as it can handle many tasks by using different prompts, and performs well even with little task-specific data (Chung et al., 2024).
However, high-quality rewriting and summarisation data need human annotation, which is time-consuming and expensive.
Recent studies show that large language models (LLMs) can generate synthetic training data for low-resource summarisation, such as "MixSumm", a method using LLaMA3 to augment few-shot data, achieved near-supervised ROUGE scores on multiple benchmarks (Sahu et al., 2025), and "ExtraPhrase" that created pseudo-paraphrases and summaries to boost summarisation in low-resource settings (Loem et al., 2022).

3 METHODOLOGY

3.1 OVERVIEW

This project proposes a multi-task NLP model to help ADHD users organise their thoughts by three key tasks: 1) multi-label classification, 2) sentence rewriting, and 3) summarisation.
The iterative project design is illustrated in Figure 1, and each step is discussed subsequently.

Figure 1: Iterative Project Design

3.2 DATA COLLECTION

The dataset was collected from the r/ADHD subreddit (2.1 million members), the largest online ADHD community, covering posts and comments from 2008 (launch date) to the end of 2024. These data were downloaded using the Arctic Shift: Photon Reddit Tool (Arctic Shift, 2025).
Table 1 shows the data size progression across different preprocessing stages. Table 2 shows the Exploratory Data Statistics results on the data, which is helpful for determining the minimum and maximum character length as the filtering criteria. Table 3 shows the data source for each task with justification.

Table 1: Raw data size progression

Stage	Posts	Comments
Raw	2.4GB (937,818 rows)	12GB (8,342,698 rows)
Filter & deduped: classification	28.6MB (341,027 rows)	2.52GB (7,363,715 rows)
Filter & deduped: rewrite & summarise	569MB (548,574 rows)	
Deduped	NA	2.50GB (7,145,753 rows)
Formatted	Classification: 31.1MB Other: 574.4MB	2.50GB (7,145,617 rows) Removed bot content 136 rows

Table 2: Exploratory Data Statistics (EDA)

Type	Min	Max	Mean	Median	90 %	95 %	99 %
Posts_Title	10	309	65.36	55	110	140	231
Posts	22	39929	1046.22	794	1940	2558	4601
Comments	1	13741	341.25	203	773	1104	2128

*Unit = character.

Table 3: Data Source and Task Pair

Task	Dataset	Reason
Classification	Posts (Title-only and Full Post), Long Comments	Model to train and generalise on various formats and content across comments and posts
Rewrites	Full Posts (Title & body), Long Comments	Long comments and full posts have richer context and structure to rewrite for clarity
Summarisation	Full Posts	Full post is rich for summarisation to extract the main idea

3.3 DATA CLEANING AND SAMPLING

To ensure data quality for the project, data pre-processing was performed as below.

a) Select mandatory column: post id, title, selftext column from post, while comment id, body column from comment.
b) Split post into two datasets: title-only posts and full posts (title and body).
c) Removed bot-generated entries, empty post.
d) Deduplication of data by title and content.
e) Formatted full posts as "text: [TITLE]: title\n[BODY]: selftext", title-only post as "text: [TITLE]: title", while comments as "text: body. "
f) Applied character limits after EDA (minimum 100, maximum 1200 for full posts, 1000 for comments).
g) Using reservoir sampling to sample 2000 entries in batches, with tracking of sampled IDs to prevent overlapping between batches. Exact number is presented in Table 4 for each data type.

Table 4: Sampled entries ratio

Type	Number of Entries	Percentage
Full posts	860	43%
Comments	860	43%
Title-only	280	14%
Total	2000	100%

h) Removed URLs, emojis, extra whitespace, and characters that FLAN-T5 AutoTokenizer could not decode from the sampled entries.

3.4 ADHD LABELLING SCHEMA

Table 5 shows the label schema in this project, defining each label and the labels exclusivity rules.

Table 5: Label Schema

Label	Definition	Rules/Exclusivity
Coping	Intentional strategies for managing ADHD-related difficulties.	Exclude if merely asking for tips
Self Reflection	Personal introspection.	Mutually exclusive with Relationships
Positive Moment	Personally meaningful experiences of success or joy.	Mutually exclusive with all labels
Health	ADHD-related symptoms, diagnosis, or treatment experiences.	Can coexist
Daily Life	Day-to-day routines and practical life experiences.	No deep reflection
Relationships	Social interactions or interpersonal experiences are influenced by others.	Mutually exclusive with Self Reflection
Work	Job-related experiences or professional challenges.	Can coexist
Study	Academic experiences or challenges in educational settings.	Can coexist
None	Generic, impersonal, or non-ADHD-specific content.	Exclusive, used when others do not fit

3.5 ANNOTATION WORKFLOW

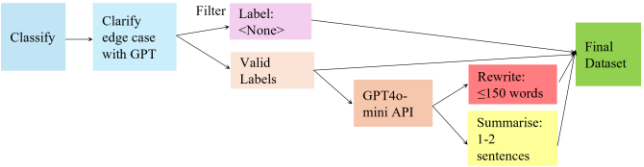

Figure 2: Annotation Workflow

1. Initial Classification: Each sample was manually classified according to the predefined labelling schema (refer Section 3.4). GPT-4o is used to discuss and clarify the ambiguous entries. Output: <LabelA>|<LabelB> for multi-label format.
2. Filtering for Subsequent Tasks: Samples classified under "<None>" were excluded from both rewrite and summarisation tasks. The dataset distribution after the initial 2000 raw data labelling is shown in Table 6.

Table 6: Dataset Distribution

Tasks	Full posts	Comments	Title-only	Total
Classified	860	860	280	2000
Rewritten	714	572	–	1286
Summarised	714	–	–	714
Total				4000

3. Prompt-Based Generation Target: GPT4o-mini was selected for rewriting and summarisation after comparison with GPT4.1-mini and GPT4o. GPT4.1-mini tended to over-interpret or hallucinate content, while GPT4o produced outputs that were too formal for the intended task.
4. The prompts were designed to align with downstream model training requirements:
5. Rewrite: Limited to 150 words, preserve original meaning and tone.
6. Summarise: 1-2 sentences, preserve only the core idea in third-person narrative referring to the user.

3.6 DATA PREPARATION

The data is split into training, validation, and test datasets by a ratio of 70:15:15. This split is done by stratified sampling by label occurrence and falls back to random sampling if the label groups are too rare. The data has been grouped by ID to ensure same data would not have different tasks in different splits (i.e., an ID is inside train_label, but in valid_rewrite or test_summary). The design is to ensure that no data leakage and impact the outcome. Table 7 shows the exact number of each split per task.

Table 7: Dataset Split

Tasks/ Split	Train	Valid	Test	Total
Classification	1400	300	300	2000
Rewrite	901	204	181	1286
Summarisation	519	103	92	714
Total	2820	607	573	4000

Upon accessing the classification results, the data is highly skewed (<Health> and <None> by 30% and 27.3% in the dataset), which leads to the class imbalance issue. This will affect the training results and lead the model to generate these frequent labels, hence oversampling is applied. The process was done by calculating the weightage of each label and multiplying the data by the weightage, with the rarest label <Positive Moment> getting 5 times higher.
After the oversampling, the train_label grows from 1400 to 2690 and is saved as train_label_balanced, and the changes are illustrated in Table 8. Note that the count is by label occurrence, so if a sample has multiple labels, all labels are included in the calculation of weightage and oversampling.

Table 8: Training classification balanced

Label	Weightage	Before	After
<None>	1.5	390 (27.9%)	390 (14.5%)
<Coping>	2	174 (12.4%)	420 (15.6%)
<Self Reflection>	2	298 (21.3%)	692 (25.7%)
<Positive Moment>	5	31 (2.2%)	155 (5.8%)
<Daily Life>	3	99 (7.1%)	299 (11.1%)

<Health>	1.5	434 (31.0%)	620 (23.1%)
<Work>	5	55 (3.9%)	275 (10.2%)
<Study>	5	54 (3.9%)	270 (10.0%)
<Relationships>	3	88 (6.3%)	276 (10.3%)

3.7 MODEL SELECTION AND TRAINING

The chosen model is FLAN-T5, a versatile multitasker model built on T5 (Text-To-Text-Transfer-Transformer) that supports instruction fine-tuning. To ensure reproducibility, this project uses a fixed 42 random seed in all scripts.
For the pilot phase, the lightweight FLAN-T5-Small model was used to develop and validate the framework. After the data preparation stage, training is continued with FLAN-T5-Base to improve performance. As T5 is a text-to-text model, limiting output to a fixed set of labels is not feasible without post-processing rules, and the model tends to bias toward high-frequency labels. Therefore, a classifier head is added to the model, predicting the labels statistically by logits.
To further enhance performance, training was then conducted on Colab Pro and supported by a 2TB Google Drive storage upgrade to accommodate model checkpoints and logs. Due to the resource constraints, the training was later migrated to Kaggle. Initial training attempted with multi-task encountered multiple issues, including out-of-memory errors, unstable loss (NaN), and task interference. To simplify the setup and improve stability, the training was restructured into staged training.
The model training flow is illustrated in Figure 3 and explained later.

Figure 3: Model Training flow

3.7.1 PREPROCESS ALL DATA FILES

a) Load all JSONL data files for all tasks (classification, rewrite, summary) and all splits (train, validation, test)
b) Preprocess the data using the FLAN-T5-Base tokenizer
c) Add custom classification labels as special tokens
d) Compute label weights based on occurrence frequency (max weight = 5 for rarest labels)
e) Save the pre-processed data and weights

3.7.2 TRAIN TASKS IN STAGES

f) Stage 1 Classification: Fine-tune FLAN-T5-Base (encoder, decoder, and additional classifier head). Use focal loss on logits to address label imbalance. LRs: encoder 6e-6, decoder 5e-6, classifier head 1.5e-4. Early stopping uses micro-F1 with a threshold value and patience of 3 epochs.
g) Stage 2 Rewrite: Freeze the classifier head and train seq2seq for the rewrite task. LRs: encoder 6e-6, decoder 2e-5. Early stopping on validation loss with label smoothing of 0.1 to improve the generation output.
h) Stage 3 Summarisation: Same as Stage 2, for summary generation only with shorter output tokens allocated.

3.7.3 ENHANCE CLASSIFICATION

a) Stage 4 Polishing: Freeze T5 component, train only the classifier in 6 epochs. Evaluate and use micro-F1 to early stops with patience 3. Final learning rate after several tries is 1e-3.

3.8 MODEL INFERENCING

Many transformer-based models, such as BERT or RoBERTa, are commonly used for classification tasks, whereas T5, as a text-to-text model, is designed for text generation. To adapt it for classification, several actions were initiated. These include monitoring F1 score during multitask training to save the best model checkpoint, as described in Section 3.7. Further inference was then performed on the validation dataset using F1 score and other evaluation metrics to determine the optimal threshold value per label for filtering false positives. The optimal threshold value will be used in the evaluation and comparison analysis with the default threshold value. Finally, the same script is used to evaluate the test data.

3.9 ETHICAL AND LEGAL CONSIDERATIONS

The dataset used in this project was collected from r/ADHD subreddit, a public online forum with over two million members. All posts and comments were retrieved using the Arctic Shift Photon Reddit Tool (Arctic Shift, 2025), covering content published from 2008 to 2024. Only publicly visible text was included, and no usernames, profile metadata, private messages, or external links were stored.
This research follows Reddit's Privacy Policy, which permits public content to be accessed and analysed by third parties, including for academic purposes. As the study does not involve any user interaction, private content, or identifiable personal data, it qualifies as a secondary analysis of public data. All processing was conducted while ensuring user privacy and adhering to the platform's terms of use.

3.10 EVALUATION STRATEGY

Due to the multi-task complexity, evaluation is performed separately for each task. As mentioned in Section 3.5, the rewritten texts and summaries were generated using GPT-4o-mini and therefore do not serve as gold-standard references. These two generative tasks are evaluated using general similarity metrics only.
The evaluation metrics are:

a) Classification: micro/macro F1 score, recall, precision, and hamming loss for multi-label classification.
b) Rewrite: BLEU (Bilingual Evaluation Understudy) and BERTScore to assess LLM-paraphrased text.
c) Summary: ROUGE (Recall-Oriented Understudy for Gisting Evaluation) and BERTScore to assess the quality of LLM-generated summaries.

4 RESEARCH FINDINGS & ANALYSIS

4.1 OVERVIEW

This chapter discusses the evaluation of all tasks designed for the multi-task trained FLAN-T5 model. It will assess the part 1 staged training, then part 2 stage 4, which enhances the classifier, with both validation and test data results presented. Note that the per-label thresholds and the mutually exclusive labels <Self Reflection> versus <Relationships> margin have been calculated on the validation split. These values have been used to evaluate the same validation split and then applied to evaluate test data. As the generation tasks were not tuned on the validation split, this paper will report generation metrics for the test split only.

4.2 STAGE 1-3 STAGED TRAINING RESULT

Table 9: Classification result with default threshold 0.5

Dataset	**Validation**				
Metrics	Micro-F1	Macro-F1	EM (Exact Match)	Hamming Loss	Weighted-F1
Scores	0.4371	0.3597	0.2033	0.1841	0.4830

Table 9 shows the classification scores before applying per-label threshold tuning or rules.

A fixed threshold of 0.5 is used for all labels, the same as in model training. The overall performance is relatively low, especially on macro-F1 and exact match, suggests the model struggles with less frequent labels and predicting full label groups correctly. This result serves as the baseline before any tuning or improvements.

Table 10: Stage 1-3 – Classification result for validation data post-rules

Dataset	**Validation**				
Metrics	Micro-F1	Macro-F1	EM	Hamming Loss	Weighted-F1
Scores	0.5189	0.4332	0.3533	0.1367	0.5495

Table 11: Stage 1-3 – Classification result for test data post-rules

Dataset	**Test**				
Metrics	Micro-F1	Macro-F1	EM	Hamming Loss	Weighted-F1
Scores	0.4980	0.4074	0.3633	0.1419	0.5142

Table 10 above represents the final classification results on the data after per-label thresholding tuning, along with the softened label exclusivity rules and the margin computed between the <Self Reflection> and <Relationships> labels. Compared to before threshold-tuning and margin between labels, the score has shown great improvement. For example, macro-F1 increased from 0.3597 to 0.4332, and exact match increased from 0.2033 to 0.3533 on the validation set.

The test results in Table 11 also show similar performance, with a macro-F1 score of 0.4074 and micro-F1 of 0.4980, indicating that the model can generalise to unseen data. Note that the classification results have shown better results on the validation split, as the threshold and margin were optimised on the same split.

Table 12: Classification result per-label for test data in stage 3

Label	**F1**	**Precision**	**Recall**	**Support**
<None>	0.7021	0.7021	0.7021	94.0
<Self Reflection>	0.5276	0.4343	0.6719	64.0
<Health>	0.4865	0.5217	0.4557	79.0
<Coping>	0.4675	0.4615	0.4737	38.0
<Work>	0.4615	0.7500	0.3333	9.0
<Relationships>	0.3529	0.4286	0.3000	20.0
<Daily Life>	0.3284	0.2558	0.4583	24.0
<Positive Moment>	0.2069	0.1154	1.0000	6.0
<Study>	0.1333	0.3333	0.0833	12.0

* Sort by F1 in descending order

Precision represents how correct the true positive prediction is, recall represents how much of the true positive is identified, and F1 represents the balance between both metrics. Support represents the number of samples for that label in the test split. Per-label performance on the test dataset shows varied results in Table 12 above.

In Table 12, <None> has balanced performance across all metrics by 0.70. Next, <Self Reflection> shows high recall (0.67) but lower precision (0.43), indicating that the model is overpredicting this label. <Coping> and <Health> have moderate scores, both around 0.46–0.52 precision and 0.45–0.47 recall, suggesting the model can detect them but not consistently.

Interestingly, <Positive Moment> reaches perfect recall (1.00) but very low precision (0.11), showing many false positives. This may be caused by the post-processing rule that forces it to be the only label when predicted and reach the optimal threshold.

<Daily Life> and <Relationships> have low to medium scores across all metrics, with F1 below 0.36. <Work> shows high precision (0.75) but low recall (0.33), meaning the model only predicts it when very confident but misses many real examples.

<Study> is the weakest label with 0.33 precision, 0.08 recall, and 0.13 F1. This suggests the model works better on frequent labels like <None> and <Health> but struggles with rare ones like <Positive Moment> and <Study>.

Table 13: Generation Result for Test Data

Dataset	**Test**			
Metrics	Rewrite BLEU	Rewrite BERT	Summary ROUGE-L	Summary BERT
Scores	0.1093	0.4953	0.3590	0.4892

Table 13 above shows that both rewrite and summary tasks follow a similar pattern, where BERT Score is significantly higher than other metrics. This is due to BERT Score measures the semantic similarity between model output and reference text, rather than relying on exact word matching. In contrast, BLEU and ROUGE-L depend more on surface-level overlap, where BLEU measures n-gram (exact word overlap) or ROUGE, which measures the occurrence of exact words in the same order. These metrics may score low if the model uses different wording or sentence structures, even when the meaning is still the same.

For example, the rewrite task achieved a BLEU score of only 0.1093, while the BERT Score reached 0.4953. This difference suggests that although the word choices may differ, the core meaning is often preserved. A similar pattern is observed in the summary task, where ROUGE-L is 0.3590 and Summary BERT reaches 0.4892, again showing that the model captures the main meaning even when surface wording differs. Earlier attempts to hyper-tune the generation parameters compared to this baseline have shown only very small improvements, which are not worth considering the extra time, cost, and resources spent.

4.3 STAGE 4 – CLASSIFIER RETRAINING RESULT

After observing unstable outputs and hallucination issues in earlier attempts at rotation multi-task training, Stage 4 was restructured to freeze the entire generation components to focus on retraining the classifier head. This modification aimed to improve classification accuracy while preserving the stable generation outputs from Stage 1–3. Earlier experiments with hyperparameter tuning for the generation tasks, before freezing, showed only very minor improvements over the baseline, which were not worth the extra time, cost, and resources.

Table 14: Stage 4 – Classification result for validation data post-rules

Dataset	**Validation**				
Metrics	Micro-F1	Macro-F1	EM	Hamming Loss	Weighted-F1
Scores	0.6137	0.5475	0.4700	0.1026	0.6362

Table 15: Stage 4 – Classification result for test data post-rules

Dataset	**Test**				
Metrics	Micro-F1	Macro-F1	EM	Hamming Loss	Weighted-F1
Scores	0.5231	0.4479	0.4233	0.1263	0.5310

After retraining the classifier head in Stage 4, the classification results improved compared to Stage 1–3. Comparing Table 14, which shows results for validation splits in Stage 4, to Table 10, which shows Stage 1–3 results, micro-F1 increased from 0.5189 to 0.6137, macro-F1 increased from 0.4332 to 0.5475, and exact match improved from 0.3533 to 0.4700. Hamming loss became lower from 0.1367 to 0.1026, suggesting that the model made fewer mistakes in label prediction. Weighted-F1 increased from 0.5495 to 0.6362 on validation, showing an overall improvement in performance across imbalanced classes.
From Table 15, the test results also show improvement compared to Table 11 for Stage 1-3 on test split, with all metrics having higher values and a lower hamming loss of 0.1263. micro-F1 0.5231, macro-F1 0.4479, weighted-F1 0.5310, and exact match 0.4233 indicate that the model is better at predicting both common and less frequent labels. These results show that focusing on retraining the classifier improves overall classification task performance.

Table 16: Classification result per-label for test data in stage 4

Label	**F1**	**Precision**	**Recall**	**Support**
<Health>	0.5752	0.5946	0.5570	79
<Self Reflection>	0.5526	0.4773	0.6562	64
<None>	0.6550	0.7273	0.5957	94
<Work>	0.4545	0.3846	0.5556	9
<Coping>	0.4194	0.5417	0.3421	38
<Daily Life>	0.3662	0.2766	0.5417	24
<Study>	0.3529	0.6000	0.2500	12
<Positive Moment>	0.3429	0.2069	1.0000	6
<Relationships>	0.3125	0.4167	0.2500	20

* Sort by F1 in descending order

Comparing per-label F1 scores between Stage 1–3 (Table 12) and Stage 4 (Table 16) shows that some labels with low performance have improved, suggesting better recognition of underrepresented or difficult labels. For example, the F1 score for <Study> increased from 0.1333 to 0.3529, <Positive Moment> increased from 0.2069 to 0.3429, and <Health> increased from 0.4865 to 0.5752. Smaller improvements are also seen in <Self Reflection>, which went from 0.5276 to 0.5526, and <Daily Life>, which increased from 0.3284 to 0.3662.
On the other hand, some labels slightly dropped in performance. <None> decreased from 0.7021 to 0.6550, and <Coping> dropped from 0.4675 to 0.4194. The F1 score for <Relationships> also went down from 0.3529 to 0.3125, possibly due to conflicting with <Self Reflection> in some cases.
Overall, the results suggest that the classifier in Stage 4 was better at handling some labels that were previously underperforming, especially those with fewer training examples or less distinct patterns.

Generation Tasks:
Same result as Stage 1-3, as the generation part has been frozen, only retrain the classifier head.

4.4 MODEL OUTPUT EXAMPLES

This subsection will present one test data and showcase the output from all tasks, using the same prompt, label definition, and post-processing rules that apply labels exclusively based on optimal threshold and margin from validation data.

Input data
[TITLE]
How am I actually supposed to survive?
[BODY]
Classic case of constantly quitting jobs and sitting around for months in between here. I'm mid-twenties, live with my parents, and there is nothing about living in the US that is appealing to me. Even in jobs where I've enjoyed the work I get that 'brain itch' over the prospect of spending most of my waking hours for the next 40-50 years there. There is literally no job that I can imagine enjoying when I have to do it 40 hours a week, but I need to have a job because I don't want to starve and my parents patience is running thin. I guess what I'm asking is how am I supposed to do

this? Or maybe are there countries I could move to that have a friendlier system for people with ADHD? Keep in mind I don't have any degrees or certifications and I understand that severely limits my ability to move out of the country. EDIT: I WANT TO WORK, I never said that I didn't. I should have clarified that I'm seeking advice on how to deal with working within a system that alienates you from your work no matter what it is you are doing.

Outputs:
Labels
<Self Reflection>, <Work>

Rewritten text
I'm mid-twenties and live with my parents in the United States, experiencing a "brain itch" due to the prospect of spending most of my time in the US for the next 40-50 years. While I enjoy the work, I need a job to avoid starvation and lack the patience of my parents, which limits my ability to move out of the country. I want advice on how to deal with working within a system that alienates me from your work, regardless of what you're doing.

Summary
User is seeking advice on how to survive in the US due to a lack of jobs and a desire to work, despite having no degrees or certifications.

4.5 ERROR ANALYSIS

To understand the model performance in the classification task, a label confusion heatmap was plotted using the test set, as shown in Figure 4 below. Each row (X-axis) represents the true label, and each column (Y-axis) represents the predicted label. Note that the values are normalised, so each row sums up to 1.

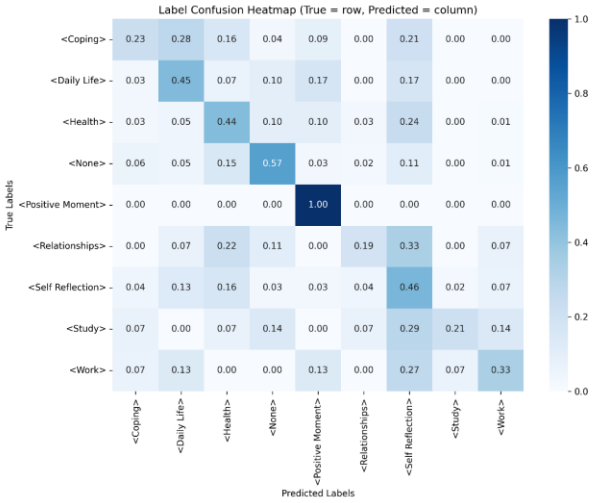

Figure 4: Label Confusion Heatmap for Classification (normalised)

The top priority label <Positive Moment> was always predicted correctly with a score of 1.00. As explained in Section 3.4, <Positive Moment> is an exclusive label that cannot co-exist with other labels. So, if the model predicts this label with high confidence, all other labels are removed by the post-processing rule.
Next, <Self Reflection> and <Relationships> are also mutually exclusive. <Self Reflection> is used for personal introspection, while <Relationships> involves interaction with other people. A post-processing rule was set to remove the weaker label if both appear.
Between these two labels, <Self Reflection> is a stronger label that has 0.46 cases predicted correctly, with 0.16 wrongly classified as <Health> and 0.13 mislabelled as <Daily Life>. Conversely, <Relationships> is the weakest label as the model only predict correctly in 0.19 cases, while mislabelling 0.33 cases as <Self Reflection>. This suggests the model struggles to tell if the reflection is self-driven or influenced by social context.
Another underperforming label is <Study>, which was only predicted correctly in 0.21 cases. It was often misclassified as <Self Reflection> (0.29), <Work> (0.14), or <None> (0.14). This suggests that the model is not confident in labelling the sample as Study when other topics are present together.
<Coping> shows a similar issue, with only 0.23 correct predictions. It was often wrongly classified as <Daily Life> (0.28), <Self Reflection> (0.21), and <Health> (0.16). This suggests that when users talk about multiple topics together, the model struggles to recognise the coping strategy being described.
As for <Work>, the correct prediction was 0.33. It was often confused with <Self Reflection> (0.27), and <Daily Life> (0.13). This shows similar patterns to <Study>, where the model is not confident in labelling the weaker background label when other dominant labels are present.
Lastly, the <None> label was correct in 0.57 of the cases, but also mislabelled as <Health> (0.15) and <Self Reflection> (0.11). This might happen when a post has uncleared signals, and the model cannot decide if it relates to ADHD and can fit into other label groups or not.
In general, the model performs better for labels that have clear signals or strong rules like <Positive Moment>. However, it still struggles with overlapping or vague cases like <Coping>, <Relationships>, and <Study>, highlighting the challenge of handling complex user-generated content, especially in ADHD multi-topic context.

5 DISCUSSION

5.1 OVERVIEW

This chapter discusses the overall experience of building an end-to-end pipeline and training a multitask model using FLAN-T5 on ADHD user-generated text from the r/ADHD subreddit. The model handles both classification and generation tasks (rewrite and summarisation), with classification as the primary focus. The following subsections cover key challenges, training strategy, iterative development, and the findings observation.

5.2 KEY CHALLENGES

5.2.1 DATA

The initial challenge lies in data preparation, as the dataset is built from real user-generated content on r/ADHD, which is natively unstructured, noisy, and informal. Moreover, the nature of ADHD users introduces further complexity, as users often jump between

topics, ramble without a clear point, or present their thoughts in a disorganised way.
Note that as this project collects both posts and comments, the context varies significantly in length, formatting, semantic density, and clarity. This variety helps the model learn to support diverse ADHD user inputs but also causes generalisation difficulties due to inconsistent structure and shifting context.
The second challenge is in the annotation of the dataset, as the model requires annotated data to perform supervised learning. The initial design is to use LLM to generate all three tasks' outputs for the FLAN-T5 model by prompt engineering with the GPT-4 family. While this approach was effective for text generation tasks such as rewriting and summarisation, it proved ineffective for classification.
The unsatisfactory performance in classification is mainly due to the difficulty of multi-label prediction, where overlapping label definitions and vague topic boundaries require contextual understanding. LLMs often fail to capture the entire context and rely on keyword matching, resulting in over-labelling. After several attempts to automate multi-label annotation using LLMs with manual review, the process eventually shifted to using the LLM only to assist in clarifying edge cases during manual annotation.
For the generation tasks, GPT-4o was initially used to generate rewritten text and summaries. However, GPT-4o produces well-written but formal text, even though the original input is informal. In contrast, GPT-4o-mini generates concise and clear text while preserving the same tone. Therefore, the generation target for this project are entirely generated by GPT-4o-mini.
On the other hand, the summary outputs are revised to only 1–2 concise sentences to capture the core idea of each paragraph, and refer to the author as ‘user’, since LLMs may mix first-person and third-person narrative styles. This adjustment also ensures consistency in model training by aligning all summaries to the same narrative perspective.
Lastly, while the rewrite prompt originally included instructions for the LLM to ignore format labels such as ‘title’ or ‘body’, this part was removed before training FLAN-T5. Including these format instructions introduces unnecessary context variation, which negatively affects the model’s rewrite performance.

5.2.2 MODEL DESIGN

As introduced in the methodology, FLAN-T5 is a text-to-text model primarily designed for generative tasks such as rewriting or summarisation. Conversely, classification requires the model to evaluate each possible label and decide whether it applies. This decision-based process is fundamentally different from producing free-form output, and the text-to-text model is not originally designed for it.
The initial design was to use the T5 body to prompt the model to predict outputs on samples in free text form. However, it produced gibberish, repetitive, and unrelated texts, and required post-processing rules to clean and return only the valid labels. Hence, the classification task no longer uses the T5 body for generation and is now handled by an additional classifier head that calculates logits and sigmoid values for each label.
The training of multiple tasks on T5 is also challenging, as the model was not able to handle the variation in context, format, and length across the three input types (titles, full posts, and comments).

5.3 TRAINING STRATEGY AND ITERATIONS

5.3.1 INITIAL TRAINING SETUP

As stated in Section 5.2.2, the classification task eventually adopted a classifier head instead of relying on the T5 body. In the initial setup with the classifier head, validation was performed only on full posts, as they were used across all tasks. This approach ensured consistent input structure and longer content within the small validation set (142 entries, with the remaining 1858 for training). However, even with helper prompts to revalidate selected labels, the F1 score remained low (around 0.30).

5.3.2 ISSUES OBSERVED

After switching to the classifier head setup, the model still struggled with simpler inputs or shorter texts, especially when tested with manually written examples such as short phrases or simulated comments. This issue was not identified in the evaluation stage, as the validation split only included full posts, which are longer and richer in context. Even when the training data did include all types of input (full posts, comments, and title-only posts), the lack of diverse examples in the validation set made it difficult to evaluate the model's generalisation capability. Moreover, without a separate test set in the early setup makes it even harder to evaluate models' performance on unseen examples. These issues only arise during manual inference, where the model underperforms when given shorter or less structured inputs. To address this, the dataset was split again by task, ensuring that each task had its own training, validation, and test sets with a 70:15:15 ratio.
After the new split, the model still failed to produce reasonable outputs. Some predictions resulted in NaN values or consistently low logits (e.g., 0.05), suggesting the model was unable to decide among the labels. A deeper analysis points out that class imbalance and variation in input context, length, and format are the main issues. To resolve these issues, several adjustments were applied: stratified splitting by labels, label-based weighting to address data skew, switching to focal loss, and oversampling rare-label data to build a more balanced training set.

5.3.3 STAGED TRAINING DESIGN

The initial design treats all tasks as text-to-text generation, hence a single file train_all.jsonl is sufficient for all tasks shared. After switching to a classifier head for the classification task, it introduced a different architecture that is not compatible with the decoder-based setup used for generative tasks. As a result, the dataset had to be split into task-specific files (train_classification, train_rewrite, and train_summary), which then led to errors such as padding issues or missing decoder input IDs when trying to train all tasks in the same loop. The model was unable to handle this variation within a unified training design.
Afterward, the model was revamped to staged training, which turns the original mixed training into four stages: Stage 1 classification, Stage 2 rewrite, Stage 3 summary, and Stage 4 rotational multitask training. In stages 1 to 3, each task is trained in isolation with its dataset. The training will freeze the T5 body during classification (Stage 1) and freeze the classifier head during generative tasks (Stages 2 and 3) to avoid task interference. Each task also applies its learning rate, number of epochs, and

weightage loss, ensuring sufficient time is allocated for the model to learn different tasks gradually.
Moving to Stage 4, a rotation schedule is introduced where the model alternates tasks per epoch, starting with 1 epoch per task, then changing to a 3:1:1 and 2:1:1 ratio (classification > rewrite > summary). Note that focal loss was also used in classification, and the classification loss was multiplied by 2, since it remains the main focus of the model training.
Later evaluation shows a significant improvement for classification with micro-F1 0.6 and macro-F1 0.53, but a low score for generation tasks (BERTScore of 0.3-0.33, BLEU 0.032, and ROUGE 0.25). Upon manual inferencing with dummy data, some critical issues are detected in the generation output, which include hallucination, pronoun mixing, and inconsistent or wrong information.
These findings lead to an observation: rotating tasks seems to degrade model performance, rather than enhancing it. Mixing up of pronouns may be due to the prompt in task 3 summarisation, where the prompt requires the model to use third-person narrative, and the contrast in summary length versus rewrite length confuses the models. Therefore, the design of the rotation task in Stage 4 was then revised to a single classification task.
During earlier experiments with the small full-post-only dataset, the helper prompt showed useful behaviour by revalidating predicted labels. However, as the dataset expanded to include a mix of post titles, comments, and full posts, the prompt became unstable. This suggests that while it worked under controlled input, it does not generalise well without fine-tuning or architectural adjustments, so the helper prompt is removed.

5.3.4 FINAL ADJUSTMENTS

Early result shows that the validation and test datasets give similar results, suggesting that the model's behaviour is stable. However, the issue remains that upon evaluation, the F1 scores are not satisfactory. This raises a question of whether the evaluation loss during training can reliably determine the best models, especially in earlier runs that used task rotation. Several options were explored to improve the F1 score, including adjusting the number of training epochs, enabling early stopping, and resorting to the final solution: using validation micro-F1 during training to decide when to save the model. A patience level of 3 with a 0.003 threshold is now used.
In addition to this, the encoder, decoder, and shared components are fully frozen in Stage 4 to preserve the generation task performance. Stage 4 utilises the classifier head only for training. The same parameters are now applied and consistent for the classification task (Stages 1 and 4).
Also, another experiment was conducted by hyper-tuning the generation task parameters and allowing more epochs aggressively, but after 4× hours taken the metrics showed only a little improvement. Considering the GPU hours and training time required, it wasn't worth it, hence the generation task was reverted to the original setting.

5.4 FINDINGS & OBSERVATIONS

5.4.1 LABEL ANNOTATION PHILOSOPHY

The annotation process in this project focuses on the classification task, where labels were designed carefully, and the data is manually annotated to guide the user in recognising the topic. For example, if a user is reflecting upon their actions but other parties are involved (such as conflict, comments, or fights), the data will be classified as <Relationships> instead of <Self Reflection>.
As for <Daily Life>, this label represents the daily life moments of ADHD users, including both calm and chaotic situations. For example, missing an appointment, forgetting something important, and multitasking at the same time are categorised as <Daily Life> instead of <Health>, even though these may look like symptoms. The fundamental concept in classifying the data is to guide the user to recognise "what happens" instead of assigning labels purely based on keyword matching.
Consequently, the exclusion rules apply to ensure that the label schema in Section 3.4 is implemented. This approach further complicates the task of labelling ADHD user-generated text that contains multiple topics at once, as it introduces ambiguity during training and evaluation.
As a result, the model tends to favour broader or more frequent labels when it cannot clearly distinguish between topics. Weaker or overlapping labels like <Study>, <Relationships>, and <Coping> are often overpowered by more dominant labels such as <Self Reflection>, <None>, or <Health>. This affects how well the model can surface less obvious themes in user input, as it often misses the weaker signals in the data.

5.4.2 GENERATION TASK OBSERVATIONS

As stated in Sections 4.2 and 4.3, ADHD individuals' text is messy, incoherent, and juggles topics due to impairments in speech, language, and attention. Hence, the generation tasks were introduced to rewrite the text clearly and concisely to help communication and summarise it into one or two sentences that capture the core idea.
The reported generation task performance in Section 4.2 shows a low score in ROUGE and BLEU but a moderate BERTScore, suggesting that while the model didn't use the same word (n-gram overlap), the meaning is preserved. However, these generation tasks show limited improvement through hyperparameter tuning. This unexpected little improvement is likely due to the strong pretraining of the FLAN-T5 model, which is already well-optimised for general instruction-based generation tasks such as rewriting and summarisation. Moreover, the dataset used in this study is relatively small, which makes it difficult to fine-tune the model further or achieve noticeable gains. Considering the little improvement over aggressive hyperparameter tuning, this project reverted to the baseline generation setting.

6 LIMITATIONS

6.1 PLATFORMS AND RESOURCE CONSTRAINT

This project is developed using Google Colab, Kaggle, and Google Drive. Initial training and inferences are carried out with a Colab Free account, followed by a Colab Pro 1-month subscription and 200 Computing Units top-up. The subscription allows access to better GPU L4 (24GB memory) /A100 (40GB memory) and a longer, more stable runtime than the standard T4 GPU on Colab Free. The development included a subscription with Google One for more storage (first month 2TB free trial, then subscribed monthly 100GB) to save all checkpoints, model versions, and outputs.
After the Colab Pro subscription ended, the project was migrated to Kaggle utilising T4 and P100 GPUs, which took a significantly

longer time for the training. Using Kaggle for free GPU hours has introduced additional limitations, including a 30-hour weekly limit on GPU hours and a 20GB working directory limit. Therefore, development on Kaggle has encountered OOM (Out of Memory) issues with T4 GPU/P100 (both 16GB memory), demanding the optimisation of memory usage.
The memory issue is especially crucial for a multistage training, hence the dataset is processed into Arrow format and uploaded to Kaggle for the model to load directly. Also, the training parameters required some adjustments, including a reduction of batch size from 8 to 4, data loader workers limited to 1, and constantly cleaning up memory after every stage's completion. These changes have allowed the training to be conducted overnight within the 9-hour limit per session by committing a new version, and upon completion (or failures), Kaggle will email to notify.
In the final week, the GPU hours at Kaggle have exceeded the limit, so the stage 4 training, model evaluation, and inferences are done with a Colab Free account.

6.2 MODEL CHANGES AND COMPARABILITY

The pilot phase with 150 samples has only been developed using FLAN-T5 small (77 million parameters), then upgraded to base (248 million parameters) after completing the annotation of the dataset. Upon investigating the generation output quality, the model was upgraded to large (783 million parameters) and observed a better quality of generation tasks. However, given the data size (4000 samples across all tasks), the overfitting risk, the future deployment cost/latency, the project reverted to the base model to balance the quality with efficiency.

6.3 DATA QUALITY, ANNOTATION, AND LABEL BIAS

The subreddit r/ADHD provides a large and active community of users sharing their experiences with ADHD. While Reddit is widely used and includes diverse age groups, it still represents a specific type of online environment. As users are self-identified and not clinically verified, this may introduce variability in how ADHD is expressed.
Additionally, ADHD is a neurodiversity group, every individual may experience it differently in terms of symptoms, language, and how they describe their daily life. The variability adds complexity to model design, as the way ADHD is expressed can vary widely across individuals. Note that Reddit is primarily English-speaking and requires basic digital literacy to sign up and share their opinion, hence it might not represent non-English speakers or those less engaged with online platforms.
As for the manual annotation of labels, the completion of 2000 samples took around 3 weeks, which is a bottleneck for a bigger sample size. Yet, the designs of the label, refining the label definitions and rules, and reviewing earlier batches to ensure consistency took another 1 week, bringing the entire process to 1 month more.
This project employs random sampling after removing bot/spam messages, with a character-length limit applied, and no filtering by context. As a result, the data is highly skewed toward the <Health> topic, discussing symptoms, medication, and health concerns. This skew naturally reflects the dominant themes discussed within the r/ADHD subreddit, where health-related concerns are frequently raised.
During annotation, some samples did not fit any label definition and were categorised as <None> as a fallback. The <None> label is mainly for: 1) third-person narrative with no personal experience, 2) the author is not ADHD, 3) sharing of general information, 4) out-of-scope content. About 23% of items are <None>, mainly from title-only posts or comments.

6.4 TOKEN LIMIT AND LLM-GENERATED TARGETS

The initial selection of samples is based on the character limit, but upon adding the prompt with the data, it goes over the input limit of T5 caps at 512 tokens. After the data cleansing and preparation, some characters are unable to be recognised by model as well. To address these issues, the auto-tokenizer from T5 was used to filter unknown characters and calculate the token size to truncate over-limit text. The total sample grows from 2000 to 4000 after filtering out <None> labels for subsequent rewrites and summary, and 3.38% (135 samples) from full post were truncated.
In addition to the token limit, the generation task has some limitations in the creation and quality of target outputs. Summary and rewrite examples were generated using ChatGPT 4o-mini, as its tone is more aligned with informal user text. These outputs were not human-rated, and only a rough manual scan was done, so quality claims are limited.

6.5 COST

Some expenses were incurred, which include: Colab Pro + 200 CU (USD 33) , 2 months Google One storage subscription (5 USD), and USD 5 for purchasing the OpenAI API key to produce generation targets. These costs aggregate to around 45 USD.
The costs reflect decisions made during development, including platform choices and model selection. A larger model (FLAN-T5-Large) was tested and showed better generation quality due to its stronger capacity but requires better GPU. Given the small data size, overfitting risk, and GPU cost, the project reverted to the base model to keep training more efficient and affordable.

7 CONCLUSION AND FURTHER WORK

This project demonstrates a designed multi-task FLAN-T5 model that assists individuals with ADHD in classifying, rewriting, and summarising their text. Utilising user-generated text from r/ADHD, the model delivers the designated tasks and achieves a reasonable score: micro-F1 0.5231, macro-F1 0.4479 for the primary classification task after fine-tuning and post-processing rules. Some weaker labels like <Coping>, <Study>, and <Relationships> are overpowered by dominant labels <Health> and <Self Reflection> with a distinct pattern and more samples, representing the data skew in nature among ADHD communities. Whereas for the generation task, the model scored a low ROUGE and BLEU but a moderate BERTScore, indicating that the model can preserve the semantic meanings of the text even when the choice of words differs.
Moving forward, several possible future directions may be considered for transforming the current POC into a practical application. Firstly, expand the dataset with more high-quality human-annotated data. This includes verified classification multi-labels, rewritten sentences, and reference summaries. The

annotations should undergo a formal human evaluation process beyond the automated metrics reported in this study. With an inter-rater agreement among human annotators, this could ensure that the data quality is consistent.

Additionally, collaborating with psychologists could help refine the label definitions, particularly for cases where distinctions are subtle or context dependent. For example, the current distinction between <Self Reflection> and <Relationships> labels is based on whether the reflection is self-initiated or triggered by interactions with others. With expert input, this project could strengthen the decision boundary between labels to distinguish the topics.

Once the dataset is expanded, the model may be scaled up to a larger architecture such as FLAN-T5-Large. The larger parameters enable the model to better handle longer and noisier text, which is essential for the designated tasks.

Also, the model could provide multi-lingual usage, increasing accessibility for ADHD users worldwide. To support more languages, the data source shall expand beyond Reddit, as it is a primarily English online platform.

To improve accessibility and enable user testing, the model could be integrated into a web or mobile application with user interface (UI). This will allow easier feedback collection for model refinement, enabling more features beyond the designated tasks. For example, adding the speech-to-text feature, saving output, sharing output with others for communication, and extracting to-do items to set up reminder.

Following user interaction, ethical and privacy concerns must be addressed. The system shall ensure secure data storage, anonymisation of user input, and full user control over submitted content. Any training on user data requires explicit consent, with clear opt-in and opt-out options presented.

Lastly, future deployment should consider usability testing and interface design that minimises cognitive load. Collaboration with UI designers alongside psychologists could ensure a positive user experience for neurodiverse users, particularly individuals experiencing sensory overload issues.

ACKNOWLEDGMENTS

This research project was completed independently, with the support of AI tools such as Cursor and ChatGPT for code generation, idea structuring, and model configuration. For generation task targets, the GPT-4o-mini was used to produce output samples, which were then reviewed and adjusted to ensure alignment with the project objectives.

The direction of this project was also shaped by my personal experiences. Exploring the challenges faced by neurodivergent individuals in communication and thought organisation has been both meaningful and motivating throughout this research.

Special thanks go to the r/ADHD Reddit community for the publicly shared content that made this research possible. Tools and platforms such as Google Colab, Kaggle, and Google Drive provided the computing resources and storage that supported development and testing across multiple stages of the project.

Appreciation is also extended to the Master of Data Science program at South East Technological University (Carlow), where I gained the foundational knowledge in machine learning, NLP, and deep learning that inspired the direction of this project.

Finally, I thank my supervisor for the initial approval and direction, which gave me the space to explore and independently shape this model-focused project under the MSc in Data Science framework

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